

Pierce County Assessor-Treasurer Data Dictionary

House Price Prediction - Data Reference Guide

Solution Engineering Project

2026-05-10

Table of contents

1	Introduction	3
1.1	Dataset Overview	3
2	1. Tax Account Table	3
2.1	Field Definitions	3
3	2. Appraisal Account Table	7
3.1	Field Definitions	8
4	3. Improvement Table	13
4.1	Field Definitions	13
5	4. Improvement Built-As Table	15
5.1	Field Definitions	16
6	5. Improvement Detail Table	22
6.1	Field Definitions	22
6.2	Common Detail Types & Examples	23
7	6. Land Attribute Table	23
7.1	Field Definitions	24
7.2	Attribute Categories	24
7.3	Common Attributes	24
8	7. Sale Table	25
8.1	Field Definitions	25

8.2	Critical Notes for Price Prediction	28
9	8. Seg/Merge Table	29
9.1	Field Definitions	29
9.2	Usage for Price Prediction	30
10	9. Taxpayer Table	30
10.1	Field Definitions	31
11	10. Tax Description Table	31
11.1	Field Definitions	32
11.2	Usage	32
12	Data Quality Summary	32
12.1	Coverage by Table	32
12.2	Missing Data Patterns	33
12.2.1	Geographic Coordinates (5-10% missing)	33
12.2.2	Building Details (varies by property type)	33
12.2.3	Sales Data (20-30% filtered as Invalid)	33
12.3	Data Types	34
13	Recommended Field Selection for Price Prediction	34
13.1	Primary Features (Highest Predictive Power)	34
13.2	Secondary Features (Moderate Predictive Power)	35
13.3	Derived/Engineered Features	35
14	Key Statistics & Ranges	35
14.1	Property Values (Current Year)	35
14.2	Property Sizes	36
14.3	Building Characteristics	36
15	Data Access & Update Schedule	36
15.1	Source Location	36
15.2	Update Schedule	37
15.3	Contact	37
16	Appendix: Data Mart Entity-Relationship Diagram	37
17	References & Resources	39
17.1	Data Documentation	39
17.2	Related Documents	39
17.3	External Resources	39

1 Introduction

This document provides a comprehensive data dictionary for the Pierce County Assessor-Treasurer dataset used in the House Price Prediction project. The dataset contains detailed property assessment and appraisal information for over 400,000 properties in Pierce County, Washington.

1.1 Dataset Overview

- **Source:** Pierce County Assessor-Treasurer Office
 - **Region:** Pierce County, Washington, USA
 - **Total Properties:** ~400,000 parcels
 - **Features:** 50+ attributes per property
 - **Historical Coverage:** 10+ years of sales transactions
 - **Update Frequency:** Annual (tax year basis)
 - **Access:** Public records (freely available)
-

2 1. Tax Account Table

Description: Tracks current tax and assessment account information for each property. Contains valuation data for both prior and current tax years.

Primary Key: parcel_number

Record Count: ~400,000 records

2.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Unique 10-digit identifier assigned to each property. Used to link all other tables.

Field Name	Data Type	Length	Description
account_type	Text	5	Classification of property account. Values: REAL (Real Property), STRUC (Structures), PERS (Personal Property), MOBIL (Mobile Home).
property_type	Text	5	Type of property asset. Values: ASIMP (Administrative Seg Improvement), LNDIM (Land and Improvements), MBLHM (Mobile Home), RRLEA (Railroad Leases), SAPP (State Assessed Personal Property), SARP (State Assessed Real Property), STRUC (Leased Land and/or Structure).
site_address	Text	50	Physical location address of the property. May differ from mailing address. "XXX" indicates GIS-estimated address.
use_code	Text	5	4-digit numeric code indicating present highest and best use of property for appraisal purposes.

Field Name	Data Type	Length	Description
use_description	Text	50	Text description of the numeric use code. Examples: Residential, Commercial, Agricultural, etc.
tax_year_prior	Integer		Tax year for which all prior year values apply (e.g., 2023).
tax_code_area_prior_year	Text	10	Geographic code describing taxing district combination. Subject to annual changes due to bond, levy, incorporation changes.
exemption_type_prior_year	Text	40	Type of exemption, if any, applied in prior tax year. Examples: Senior exemption, disability, church property, etc.
current_use_code_prior_year	Text	5	Coded use type in prior year. Values: AGRI (Agricultural), CT (Current Use Timber), FORDG (Forest Designated), OPBRS (Open Space PBRS), OPEN (Open Space).
land_value_prior_year	Integer		Total fair market value of land only (excluding improvements) for prior tax year, as determined by appraisal process.

Field Name	Data Type	Length	Description
improve- ment_value_prior_year	Integer		Total fair market value of all improvements/buildings for prior tax year, determined by appraisal process.
total_mar- ket_value_prior_year	Integer		Total fair market value (land + improvements) for prior tax year before exemptions.
tax- able_value_prior_year	Integer		Total market value minus exemptions granted for prior tax year. This is the value used for tax calculation.
tax_year_current	Integer		Current tax year for which all current year values apply (e.g., 2024).
tax_code_area_cur- rent_year	Text	10	Geographic code describing current year taxing district combination.
exemption_type_cur- rent_year	Text	40	Type of exemption, if any, applied in current tax year.
cur- rent_use_code_cur- rent_year	Text	5	Coded use type in current year. Same values as prior year field.
land_value_cur- rent_year	Integer		Total fair market value of land for current tax year. Used for price prediction models.
improve- ment_value_cur- rent_year	Integer		Total fair market value of improvements for current tax year.

Field Name	Data Type	Length	Description
total_market_value_current_year	Integer		Total fair market value for current tax year (PRIMARY PREDICTION TARGET).
taxable_value_current_year	Integer		Current year taxable value after exemptions.
range_code	Text	10	Geographic range designation (Township-Range-Section system).
township	Text	10	Township designation in Public Land Survey System.
section_code	Text	20	Section designation in PLSS grid.
quarter_section	Text	20	Quarter-section designation (NE, NW, SE, SW, etc.).
subdivision_name	Text	50	Name of subdivision, if property is within a recorded subdivision.
located_on_parcel	Text	10	For improvement-only parcels: the real property parcel number on which located.

3 2. Appraisal Account Table

Description: Detailed property appraisal data containing physical characteristics, land features, and valuation information. Represents professional appraiser assessment of property.

Primary Key: parcel_number (one-to-one relationship with tax_account)

Foreign Key: parcel_number references tax_account

Record Count: ~400,000 records

3.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Unique identifier. References tax_account. Links to improvement, land_attribute, and sale tables.
appraisal_ac-count_type	Text	15	Classification of property for appraisal purposes. 15 types: Com Condo, Com Leasehold, Com Multi Unit, Commercial, Industrial, Mobile Home, Reference, Res Com Condo, Res Leasehold, Residential, Trended Invest. IMPORTANT FOR PRICE PREDICTION: Residential vs Commercial distinction.
business_name	Text	50	Main business name on the parcel, or space number for mobile/manufactured home in a park. Often blank for residential.

Field Name	Data Type	Length	Description
value_area_id	Text	10	Physical inspection cycle identifier (PI1 to PI6). Indicates when property was last physically inspected. Used to assess appraisal recency.
land_economic_area	Text	30	Also known as LEA. Ties parcels to land valuation models. For Commercial: identifies land use model by PI year. For Residential: location code identifying appraisal area, sector, subsector. For Res Com Condos: relates to account land value. KEY LOCATION FEATURE.
buildings	Integer		Number of building improvements located on the parcel. Critical feature: 0 = vacant land, 1+ = improved property.
group_account_number	Text	30	For group accounts: identifies all parcels constituting one economic unit (contiguous assessment). For condominiums: identifies all units in project.

Field Name	Data Type	Length	Description
land_gross_acres	Decimal	15.4	Total acres of economic unit (may include multiple parcels). Used for large properties and commercial analysis.
land_net_acres	Decimal	15.4	Net acreage of individual parcel only. PRIMARY LAND SIZE METRIC for single-parcel properties. Range: 0.01 to 100+ acres.
land_gross_square_feet	Decimal	15.4	Total square feet of economic unit. Gross measure including non-usable areas.
land_net_square_feet	Decimal	15.4	Net square feet of individual parcel. CRITICAL FEATURE for price prediction. Typical range: 2,000 to 50,000 sq ft for residential.
land_gross_front_feet	Decimal	15.4	For commercial: front feet facing arterial street. For residential: front feet facing address street. For waterfront: effective waterfront length. Affects commercial value significantly.
land_width	Integer		Residential use only. Equal to net effective waterfront front feet. Used for waterfront properties.

Field Name	Data Type	Length	Description
land_depth	Integer		Residential use only. Net effective depth from bulkhead to inland property line. Used for waterfront properties. Measured from top of bank.
sub-merged_area_square_feet	Integer		Area of parcel that is submerged. Not included in gross or net square feet. Indicates waterfront property.
appraisal_date	Date		Most recent date appraiser physically observed property and/or made record changes. DATA RECENCY INDICATOR: Older dates suggest stale appraisals. Format: YYYY-MM-DD.
waterfront_type	Text	30	Describes type of waterfront property adjoins or has legal access to. Values: Lake, Puget Sound, River, Creek, None, etc. SIGNIFICANT PRICE PREMIUM FEATURE.

Field Name	Data Type	Length	Description
view_quality	Text	30	Market appeal of view from dwelling/property. Values: None, Fair, Good, Excellent. QUALITATIVE PRICE DRIVER.
utility_electric	Text	30	Identifies if electric power is installed, available, or not available on property. Essential utility. Values: Available, Not Available, Installed.
utility_sewer	Text	30	Sewer/septic system availability. Values: Available, Not Available, Septic system, Shared septic. Critical for residential value.
utility_water	Text	30	Water availability. Values: Available, Not Available, Well water, City water.
street_type	Text	30	Identifies if access street is paved or unpaved. Affects property access and value. Blank for reference parcels or mobile homes.

Field Name	Data Type	Length	Description
latitude	Decimal	10.5	Latitude of parcel centroid (decimal degrees). GEOGRAPHIC FEATURE for spatial analysis. Missing for ~5-10% (building-only, mineral rights, reference parcels). Range: 47.0 to 47.9 for Pierce County.
longitude	Decimal	10.5	Longitude of parcel centroid (decimal degrees). GEOGRAPHIC FEATURE. Range: -122.0 to -121.0 for Pierce County.

4 3. Improvement Table

Description: Building-level improvement data. Multiple improvements can exist on a single parcel. Contains property type classification and space metrics.

Primary Key: (parcel_number, building_id)

Foreign Key: parcel_number references appraisal_account

Record Count: ~500,000+ records (1.2x property count due to multiple improvements)

4.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Unique parcel identifier. Links to appraisal_account and tax_account.
building_id	Text	5	Unique identifier for each improvement record on a parcel. Sequential, not always continuous. Indicates which building when multiple exist.
property_type	Text	15	Classification of the building type. Values: Commercial, Duplex, Industrial, Mobile Home, Multiple Unit, Out Building, Residential, Townhouse, Triplex.
			KEY CATEGORICAL FEATURE for price prediction.
neighborhood	Text	10	Neighborhood code representing boundaries of similar social, environmental, economic, governmental forces. For Residential: same as LEA. For Commercial: defines property location.
			STRONG PRICE PREDICTOR: Neighborhood premium/discount.

Field Name	Data Type	Length	Description
neighborhood_extension	Text	10	Sub-classification within neighborhood. For Commercial: primary occupancy/use. For Residential: set to zero. For Mobile Homes: land ownership indicator (0 = owner-owned, 1-4 = park-leased).
square_feet	Integer		Sum of square feet for all 'built-as' types on building. CRITICAL SIZE FEATURE for price prediction. Typical residential range: 800 to 5,000 sq ft. Typical commercial: 5,000+ sq ft.
net_square_feet	Integer		For commercial: sum of net rentable area(s). For residential: equals built_as_square_feet or zero. Used for income calculation on commercial properties.

5 4. Improvement Built-As Table

Description: Detailed construction specifications for buildings. Records the original construction style/purpose and building characteristics. One primary record per building, additional records for additions/modifications.

Primary Key: (parcel_number, building_id, builtas_number)

Foreign Key: (parcel_number, building_id) references improvement

Record Count: ~500,000+ records

5.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Parcel identifier.
building_id	Text	5	Building identifier on parcel.
builtas_number	Integer		Sequential number identifying each built-as record. Largest built-as is number 1. Indicates primary vs. additions.
builtas_id	Integer		Numeric code for construction purpose/style. Links to Marshall and Swift cost tables. Used in appraisal valuation.
builtas_description	Text	255	Text description of construction purpose/style. Examples: "1 Story Residential", "2 Story Residential with Basement", "Office Building", "Warehouse".
builtas_square_feet	Integer		Square feet for this specific built-as type. Sum equals building square_feet.

Field Name	Data Type	Length	Description
hvac	Integer		Code for predominant heating source. Values: 1=Forced Air, 2=Baseboard, 3=Steam, 4=Radiant, etc. BUILDING SYSTEMS FEATURE.
hvac_description	Text	30	Text description of heating system. Examples: Forced Air Gas, Electric Baseboard, Steam Boiler, Radiant Heat. Affects operating costs and comfort (price component).
exterior	Text	25	Predominant exterior material on residential buildings. Values: Wood Siding, Vinyl, Brick, Stucco, Cement Board, Log, Metal. CONDITION/QUALITY FEATURE.
interior	Text	15	Predominant interior wall material (collected for residential and mobile homes only). Values: Drywall/Sheetrock, Paneling, Plaster. Indicates age/renovation status.

Field Name	Data Type	Length	Description
stories	Decimal	13.2	Number of floors above grade. Does not include attic or basement. IMPORTANT: 1 story vs 2 story is price differentiator. Values: 1.0, 1.5, 2.0, 2.5, 3.0, etc.
story_height	Integer		Building ceiling height in feet for first floor. Indicates space quality. Commercial: average of all floors. Residential: majority floor height. Typical: 8-10 feet residential, 12-14 feet commercial.
sprinkler_square_feet	Integer		Total square footage covered by sprinkler system. Zero if no sprinklers. Fire safety feature affecting commercial valuation.
roof_cover	Text	20	Roofing material. Values: Composition Shingles, Wood Shake, Concrete Tile, Slate, Metal, Rubber/EPDM. CONDITION/DURABILITY FEATURE: Newer materials (asphalt shingles 20-year, metal 40+ year) affect replacement cost.

Field Name	Data Type	Length	Description
bedrooms	Integer		Number of bedrooms in residential property. PRIMARY FEATURE for residential valuation. Typical: 2-4 bedrooms. Collected for informational purposes.
bathrooms	Decimal	7.2	Number of bathrooms (decimal format). PRIMARY FEATURE for residential valuation. Calculation: Full bath=1.0, 3/4 bath=0.75, 1/2 bath=0.5, 1/4 bath=0.25. Examples: 2.5 = 2 full + 1 half bath. Range: 0.5 to 5.0+.
units	Integer		Number of separate units within building. 1 = single-family, 2 = duplex, 3+ = multi-unit. PROPERTY TYPE INDICATOR. Affects valuation approach (per-unit vs. whole building).

Field Name	Data Type	Length	Description
class_code	Text	20	Cost class for commercial buildings. Codes for framing (wood, steel, concrete), walls, floors, roof, fireproofing. Affects construction cost estimates.
class_description	Text	50	Text description of building class. Used primarily for commercial valuation.
year_built	Integer		Year building was constructed (actual). Based on permit or historical records. CRITICAL AGE FEATURE for depreciation. Format: YYYY (e.g., 1985, 2010, 2023).
year_remodeled	Integer		Year major improvements/renovations completed. Extends building life expectancy. Impacts depreciation calculation. Zero if no remodel recorded.

Field Name	Data Type	Length	Description
adjusted_year_built	Integer		If > year_built: indicates major improvements made beyond normal maintenance (roof, HVAC, electrical, plumbing, structural). Reduces effective age. Formula: physical_age = current_year - adjusted_year_built.
physical_age	Integer		CONDITION INDICATOR. Calculated field: current_year - adjusted_year_built. Represents effective age for depreciation. Critical for price prediction. Typical: 0-80+ years.
builtas_length	Integer		Length of mobile/manufactured home in feet. Used for mobile home valuation. Typical: 50-80 feet.
builtas_width	Integer		Width of mobile/manufactured home in feet. Typical: 12-16 feet.
mobile_home_model	Text	50	Model name/number of mobile home. Used for market comparables research.

6 5. Improvement Detail Table

Description: Specific improvement features and amenities. Captures details not covered in main improvement record (garages, porches, fireplaces, etc.).

Primary Key: (parcel_number, building_id, detail_type)

Foreign Key: (parcel_number, building_id) references improvement

Record Count: 200,000+ records (features vary by property)

6.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Parcel identifier.
building_id	Text	5	Building identifier.
detail_type	Text	10	Category of improvement feature. Values: Add On, Appliance, Balcony, Basement, Carport, Elevator, Fixture, Garage, Mezzanine, Porch, Rough In, Storage. AMENITY CLASSIFIER.
detail_description	Text	50	Specific description of feature. Examples: "Fireplace (wood)", "Garage (2-car attached)", "Deck (wood 12x16)", "Finished Basement", "3 Fixtures Bath".

Field Name	Data Type	Length	Description
units	Decimal	15.4	Quantity of feature. Either count (for discrete items) or square footage (for spaces). Examples: "Fireplace: 1 count", "Garage: 520 sf", "Porch: 200 sf". FEATURE VALUE CONTRIBUTION.

6.2 Common Detail Types & Examples

Detail Type	Examples	Typical Units
Garage	1-car, 2-car, 3-car attached/detached	Square footage or count
Deck/Porch	Wood, composite, covered, uncovered	Square footage
Basement	Finished, unfinished, partial	Square footage
Fireplace	Wood, gas, electric	Count
Appliances	Built-in microwave, range hood	Count or allowance
Bathroom Fixtures	Fixture count varies	Count or allowance
Storage	Built-in shelving, pantry	Count or allowance

7 6. Land Attribute Table

Description: Land-specific characteristics and special attributes affecting property valuation. Qualitative features that may or may not contribute to value.

Primary Key: (parcel_number, attribute)

Foreign Key: parcel_number references appraisal_account

Record Count: 100,000+ records (multiple attributes per property)

7.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Parcel identifier.
attribute	Text	30	Attribute category. Format: Property Type (R/C/X) + Category. Examples: “R-Neighborhood”, “C-Economic”, “R-Amenities”, “X-Functional”. CATEGORICAL FEATURE.
attribute_description	Text	30	Specific attribute value. Examples: “View Average”, “NBHD Gated”, “WF Bank Low”, “Water Available”, “Hillside Lot”, “Corner Lot”.

7.2 Attribute Categories

- **Residential (R):** Neighborhood, Amenities, Functional, Economic, Utilities
- **Commercial (C):** Economic, Functional, Neighborhood, Utilities
- **Res Com Condo (X):** Combined residential/commercial attributes

7.3 Common Attributes

Attribute	Examples	Impact
Waterfront	WF Bank Low, WF View, WF Deeded	Premium feature
Gated Community	NBHD Gated, Secured Entry	Security premium
Corner Lot	Corner Lot, Irregular	Location premium/discount
View	View Excellent, View Average, View None	View premium (20-40% impact)
Slope	Steep, Gentle, Level	Buildability factor

Attribute	Examples	Impact
Access	Deeded Beach Access, Common Area	Functional feature

8 7. Sale Table

Description: Historical property sales transactions recorded with county auditor. **CRITICAL TABLE** for model training - contains actual market prices.

Primary Key: etn (Excise Tax Number)

Foreign Key: parcel_number references tax_account

Record Count: ~50,000 annual transactions; 500,000+ in last 10 years

8.1 Field Definitions

Field Name	Data Type	Length	Description
etn	Text	11	Excise Tax Number. Unique transaction ID issued by county auditor when deed recorded. Format: 11-character recording number. Links to county deed recording. TRANSACTION IDENTIFIER.
parcel_count	Integer		Number of parcels involved in single transaction. Usually 1, can be >1 for bulk sales.

Field Name	Data Type	Length	Description
parcel_number	Text	10	Parcel number(s) involved in sale. Links to tax_account and appraisal_account. May be multiple parcels per transaction.
sale_date	Date		Date legal document (deed) was executed (YYYY-MM-DD). TRANSACTION TIMING FEATURE. Used to filter recent market data and analyze trends. Format: YYYY-MM-DD.
sale_price	Decimal	15.2	Dollar amount recorded on ETN. PRIMARY PREDICTION TARGET VARIABLE or feature for model training. Range: \$10,000 to \$10,000,000+ for Pierce County.
deed_type	Text	40	Type of document conveying interest/legal title. Values: Warranty Deed, Quit Claim, Grant Deed, Trustee's Deed, etc. Indicates transaction formality.

Field Name	Data Type	Length	Description
grantor	Text	80	Seller: Individual(s) or company conveying ownership/interest in property. May be truncated for multi-party transactions.
grantee	Text	80	Buyer: Individual(s) or company purchasing ownership/interest in property.
valid_invalid	Text	7	CRITICAL FOR MODEL TRAINING: Codes validity of sale for appraisal purposes. Values: “Valid” (open market, unrelated parties, no pressure), “Invalid” (family transfer, estate sale, forced sale, < 1% interest, etc.). IMPORTANT: Filter to “Valid” only for accurate models. Approximately 70-80% of recorded sales are “Valid”.
confirmed_unconfirmed	Text	11	Whether property characteristics and transaction circumstances verified by assessor staff. Values: “Confirmed” (staff verified), “Unconfirmed” (not verified). Indicates data confidence level.

Field Name	Data Type	Length	Description
exclude_reason	Text	30	If invalid: reason for exclusion from appraisal sales studies. Examples: "Family different last names", "Estate sale", "Forced Sale Trans in Lieu Fcl", "Improved after sale", "1% Interest", "Non-arms length". Helps identify problematic transactions.
improved_vacant	Text	8	Property improvement status at sale. Values: "Improved" (has building value), "Vacant" (vacant land, no building). Distinguishes between land and improved property sales.
appraisal_ac-count_type	Text	15	Property classification at time of sale. Same values as appraisal_ac-count.appraisal_ac-count_type. Indicates if residential, commercial, etc.

8.2 Critical Notes for Price Prediction

- **Filter for Valid Sales Only:** Remove ~20-30% marked "Invalid"
- **Use Recent Data:** Last 5-10 years most relevant to current market
- **Adjust for Inflation:** Historical prices should be inflation-adjusted

- **Consider Seasonality:** Sales patterns vary by season
- **Data Lag:** Sales recorded 1-2 weeks after closing

9 8. Seg/Merge Table

Description: Property segregation and merger history. Tracks when parcels are split (segregated) or combined (merged). Important for understanding parcel history and long-term ownership patterns.

Primary Key: segmerge_number

Foreign Key: parcel_number references tax_account

Record Count: 50,000+ transactions

9.1 Field Definitions

Field Name	Data Type	Length	Description
segmerge_number	Text	12	Unique number assigned to segregation event. Identifies the transaction.
parent_child_indicator	Text	1	Relationship indicator. Values: "P" (Parent - inactivated parcel being replaced), "C" (Child - new parcel created from segregation). Can be multiple parents or children per segregation.
parcel_number	Text	10	Parcel involved. May reference parent parcel (being split) or child parcel (newly created).

Field Name	Data Type	Length	Description
continued_indicator	Text	1	Post-segregation status. Values: “Y” (parcel still active after segregation), “N” (parcel inactivated). Parent parcels often marked “N” when split into children.
completed_date	Date		Date segregation was completed. Format: YYYY-MM-DD. Indicates when split became official.
tax_year	Text	4	Tax year of segregation. Example: “2023”. Used to track timeline.

9.2 Usage for Price Prediction

- Track property history (same parcel through multiple transactions)
- Identify new development parcels (recently segregated)
- Detect land consolidation (mergers for investment/development)
- Account for parcel number changes in longitudinal analysis

10 9. Taxpayer Table

Description: Property owner/taxpayer information. **PRIVACY RESTRICTED** - Limited availability per Washington RCW.

Primary Key: parcel_number

Foreign Key: parcel_number references tax_account

Record Count: ~400,000 records (restricted)

IMPORTANT: This table is subject to privacy restrictions per RCW 42.56.070(9). Taxpayer names and mailing addresses are not publicly available without formal records request justification.

10.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Parcel identifier.
taxpayer_name	Text	150	Name of property owner/taxpayer. RESTRICTED -
mailing_address	Text	100	Privacy protected. Mailing address. RESTRICTED -
city	Text	50	Privacy protected. City for mailing. RESTRICTED.
state	Text	2	State for mailing. RESTRICTED.
zip_code	Text	10	ZIP code for mailing. RESTRICTED.
phone_number	Text	15	Contact phone. RESTRICTED.

Note: For this project, taxpayer information is not needed. Focus on property characteristics and market data instead.

11 10. Tax Description Table

Description: Detailed tax coding and legal descriptions for parcels. Provides text-based property description used in tax assessments.

Primary Key: (parcel_number, line_number)

Foreign Key: parcel_number references tax_account

Record Count: 500,000+ records (multiple lines per parcel)

11.1 Field Definitions

Field Name	Data Type	Length	Description
parcel_number	Text	10	Parcel identifier.
line_number	Integer		Sequential line number within tax description (1, 2, 3, ...). Multiple lines comprise complete description.
tax_description_line	Text	200	Text of tax description for this line. Legal language describing property location, bounds, and characteristics. Examples: "Lot 5, Block 2, Woodland Park Addition", "All of Section 1, Township 22N, Range 4E" (partial).

11.2 Usage

- Reconstruct full legal property description
- Identify lot/block subdivision memberships
- Track zoning boundaries
- Support legal/title work

12 Data Quality Summary

12.1 Coverage by Table

Table	Records	Completeness	Key Issues
tax_account	~400K	100%	All active parcels
appraisal_account	~400K	98%+	2% missing some fields
improvement	~500K+	95%+	Multiple per parcel
improvement_builtas	~500K+	95%+	Variable by property type
improvement_detail	~200K	70%	Only for featured properties
land_attribute	~100K	50%	Only for properties with special attributes
sale	~50K/year	100%	All recorded transactions
seg_merge	~50K hist	100%	Historical events
taxpayer	~400K	0%	Privacy restricted
tax_description	~500K	90%	Most parcels have

12.2 Missing Data Patterns

12.2.1 Geographic Coordinates (5-10% missing)

- Reference parcels (non-taxable)
- Mineral rights properties
- Special-use parcels
- Building-only improvements on others' land

12.2.2 Building Details (varies by property type)

- Vacant land: no improvement data
- Mobile homes: some fields inapplicable
- Commercial: residential-specific fields blank
- Special properties: minimal improvement records

12.2.3 Sales Data (20-30% filtered as Invalid)

- Family transfers
- Estate sales
- Forced sales/foreclosures

- Inter-company transactions
- Sales with undue pressure
- Percent interest sales

12.3 Data Types

Type	Count	Examples
Text/Varchar	35+	parcel_number, property_type, address
Integer	20+	buildings, bedrooms, bathrooms, square_feet
Decimal	10+	land_acres, land_square_feet, latitude, longitude
Date	8	sale_date, appraisal_date, completed_date

13 Recommended Field Selection for Price Prediction

13.1 Primary Features (Highest Predictive Power)

From Appraisal Account:

- total_market_value_current_year (TARGET for supervised learning)
- land_net_square_feet (primary land size)
- latitude, longitude (location)
- waterfront_type (qualitative location)
- view_quality (qualitative location)

From Improvement:

- buildings (count of improvements)
- property_type (categorical: Residential, Commercial, etc.)
- neighborhood (strong price predictor)
- square_feet (building size)

From Improvement Built-As:

- year_built, adjusted_year_built (age)
- physical_age (calculated age)
- bedrooms, bathrooms (residential features)
- stories (building size indicator)
- exterior, roof_cover (condition)

From Sale:

- sale_price (actual market prices)
- sale_date (temporal feature)
- valid_invalid (filter for quality sales)

From Land Attribute:

- attribute, attribute_description (special features)

13.2 Secondary Features (Moderate Predictive Power)

- utility_electric, utility_sewer, utility_water
- hvac_description (building systems)
- hvac (heating type)
- improvement_value_current_year (construction cost proxy)
- group_account_number (multi-parcel indicator)
- street_type (access quality)
- appraisal_account_type (property classification)

13.3 Derived/Engineered Features

Create from raw data:

- price_per_sqft = total_market_value / land_net_square_feet
- price_per_buildsqft = total_market_value / building_square_feet
- land_value_ratio = land_value / total_market_value
- improvement_value_ratio = improvement_value / total_market_value
- effective_age = adjusted_year_built - current_year
- property_age_category = binned(adjusted_year_built)
- is_waterfront = (waterfront_type != "None")
- has_excellent_view = (view_quality == "Excellent")
- improvement_premium = improvement_value - baseline_cost

14 Key Statistics & Ranges

14.1 Property Values (Current Year)

Metric	Min	Max	Median	Mean
Land Value	\$0	\$5M+	\$150K	\$250K

Metric	Min	Max	Median	Mean
Improvement Value	\$0	\$8M+	\$250K	\$400K
Total Market Value	\$10K	\$12M+	\$400K	\$550K
Taxable Value	\$5K	\$10M+	\$350K	\$475K

14.2 Property Sizes

Metric	Min	Max	Typical
Land Net Acres	0.01	50+	0.15 (residential)
Land Net Sq Ft	1,000	2,000,000+	6,500 (residential)
Building Sq Ft	100	100,000+	1,800 (residential)
Price per Sq Ft	\$10	\$500+	\$150-200

14.3 Building Characteristics

Feature	Range	Typical
Year Built	1880-2024	1990
Age (years)	0-140	30
Bedrooms	0-8	3
Bathrooms	0-6	1.75
Stories	1.0-4.0	1.5
Garages	0-3	1-2

15 Data Access & Update Schedule

15.1 Source Location

- **Website:** <https://www.piercecountywa.gov/736/Data-Downloads>
- **Format:** CSV files (downloadable)
- **License:** Public records (unrestricted use)
- **Cost:** Free

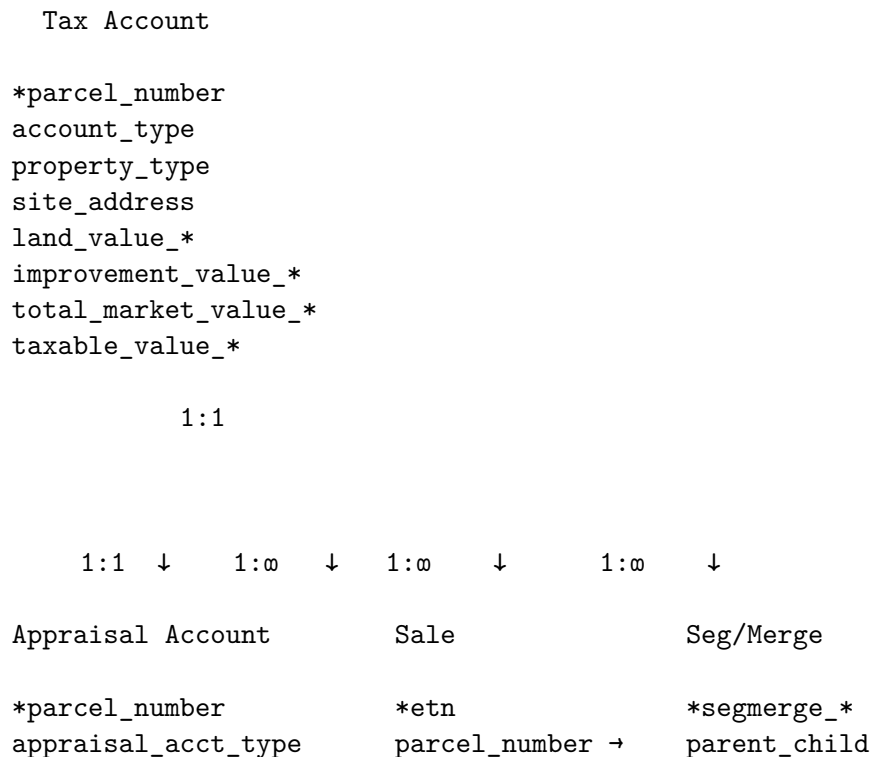
15.2 Update Schedule

- **Frequency:** Annual (tax year basis)
- **Sales Data:** Continuous (updated weekly/monthly)
- **Appraisal Data:** Rotating inspection cycles (PI1-PI6)
- **Tax Values:** Annual (based on assessor valuations)

15.3 Contact

- **Pierce County Assessor-Treasurer Office**
 - **Website:** <https://www.piercecountywa.gov/>
 - **Records Request:** Pierce County Public Records Request Center
 - **Phone:** (253) 798-7000
-

16 Appendix: Data Mart Entity-Relationship Diagram



buildings	sale_date	parcel_number→
land_net_acres	sale_price	continued_*
land_net_sqft	deed_type	completed_*
latitude, longitude	valid_invalid	
waterfront_type	improved_vacant	
view_quality		
utilities		

1:∞

↓

Improvement

*parcel_number
*building_id
property_type
neighborhood
square_feet
net_square_feet

↓

Land Attribute

*parcel_number
*attribute
attribute_*

1:∞

↓

Improvement Built-As

*parcel_number
*building_id
*builtas_number
builtas_description
square_feet
hvac, hvac_description
exterior, interior
stories, story_height
year_built, remodeled
bedrooms, bathrooms
units (# of units)

↓

Improvement Detail

*parcel_number
*building_id
*detail_type
detail_description
units

17 References & Resources

17.1 Data Documentation

- Pierce County Assessor-Treasurer Data Mart: Table Relationships and Primary Keys
- Washington State RCW 42.56.070(9) - Privacy Requirements
- Pierce County Assessment Methods and Standards

17.2 Related Documents

- PIERCE_COUNTY_DATA_MART.md - Complete data mart documentation
- PIERCE_COUNTY_SCHEMA.sql - SQL schema for implementation
- DATA_SOURCE_DESCRIPTION.md - Data source overview

17.3 External Resources

- Pierce County Official Website: <https://www.piercecountywa.gov/>
- Washington State Department of Revenue: <https://dor.wa.gov/>
- Real Estate Standards: IAAO (International Association of Assessing Officers)

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